CHAPTER

5

Short Tandem Repeat (STR) Loci and Kits

GENETIC MARKERS AND REPEATED DNA SEQUENCES

Since it has been estimated that over 99.7% of the human genome is the same from individual to individual, regions that differ need to be found in the remaining 0.3% in order to tell people apart at the genetic level. There are many repeated DNA sequences scattered throughout the human genome. As these repeat sequences are typically located between genes, they can vary in size from person to person without impacting the genetic health of the individual.

Eukaryotic genomes are full of repeated DNA sequences (Ellegren 2004). These repeated DNA sequences come in all sizes and are typically designated by the length of the core repeat unit and the number of contiguous repeat units or the overall length of the repeat region. Long repeat units may contain several hundred to several thousand bases in the core repeat.

These regions are often referred to as *satellite* DNA and may be found surrounding the chromosomal centromere. The term satellite arose due to the fact that frequently one or more minor "satellite bands" were seen in early experiments involving equilibrium density gradient centrifugation (Britten & Kohne 1968, Primrose 1998).

The core repeat unit for a medium-length repeat, sometimes referred to as a *minisatellite* or a VNTR (variable number of tandem repeats), is in the range of approximately 8 base pairs (bp) to 100 bp in length (Tautz 1993, Chambers & MacAvoy 2000). The most commonly used minisatellite marker in the 1990s was D1S80, which has a 16 bp repeat unit and contains alleles spanning the range of 14 to 41 repeat units (Kasai et al. 1990, Butler 2010, *Fundamentals*, Chapter 3).

DNA regions with repeat units that are 2 bp to 7 bp in length are called *microsatellites*, *simple sequence repeats* (SSRs), or most usually *short tandem repeats* (STRs). STRs have become popular DNA repeat markers because they are easily amplified by the polymerase chain reaction (PCR) without the problems of differential amplification. This is because both alleles from a heterozygous individual are similar in size since the repeat size is small.

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The number of repeats in STR markers can be highly variable among individuals, which makes these STRs effective for human identification purposes.

Literally thousands of polymorphic microsatellites have been characterized in human DNA and there may be more than a million microsatellite loci present depending on how they are counted (Ellegren 2004). Regardless, microsatellites account for approximately 3% of the total human genome (International Human Genome Sequencing Consortium 2001). STR markers are scattered throughout the genome and occur on average every 10,000 nucleotides (Edwards et al. 1991, Collins et al. 2003, Subramanian et al. 2003). However, not all STR loci exhibit variability between individuals.

Computer searches of the recently available human genome reference sequence have cataloged the number and nature of STR markers in the genome (see Collins et al. 2003, Subramanian et al. 2003). A large number of STR markers have been characterized by academic and commercial laboratories for use in disease gene location studies. For example, the Marshfield Medical Research Foundation in Marshfield, Wisconsin (http://research.marshfieldclinic.org/genetics/) has gathered genotype data on over 8000 STRs that are scattered across the 23 pairs of human chromosomes for the purpose of developing human genetic maps (Broman et al. 1998, Ghebranious et al. 2003).

To perform analysis on STR markers, the invariant flanking regions surrounding the repeats must be determined. Once the flanking sequences are known then PCR primers can be designed and the repeat region amplified for analysis. New STR markers are usually identified in one of two ways: (1) searching DNA sequence databases such as GenBank for regions with more than six or so contiguous repeat units (Weber & May 1989, Collins et al. 2003, Subramanian et al. 2003); or (2) performing molecular biology isolation methods (Edwards et al. 1991, Chambers & MacAvoy 2000).

Types of STR Markers

STR repeat sequences are named by the length of the repeat unit. Dinucleotide repeats have two nucleotides repeated next to each other over and over again. Trinucleotides have three nucleotides in the repeat unit, tetranucleotides have four, pentanucleotides have five, and hexanucleotides have six nucleotides in the core repeat. Theoretically, there are 4, 16, 64, 256, 1024, and 4096 possible motifs for mono-, di-, tri-, tetra-, penta-, and hexanucleotide repeats, respectively (Jin et al. 1994). However, because microsatellites are tandemly repeated, some motifs are actually equivalent to others. For reasons that will be discussed below, tetranucleotide repeats have become the most popular STR markers for human identification.

STR sequences not only vary in the length of the repeat unit and the number of repeats but also in the rigor with which they conform to an incremental repeat pattern. STRs are often divided into several categories based on the repeat pattern. *Simple repeats* contain units of identical length and sequence, *compound repeats* comprise two or more adjacent simple repeats, and *complex repeats* may contain several repeat blocks of variable unit length as well as variable intervening sequences (Urquhart et al. 1994). *Complex hypervariable repeats* also exist with numerous non-consensus alleles that differ in both size and sequence and are therefore challenging to genotype reproducibly (Urquhart et al. 1993, Gill et al. 1994). This last category of STR markers is not as commonly used in forensic DNA typing due to

difficulties with allele nomenclature and measurement variability between laboratories, although several commercial kits now include the complex hypervariable STR locus SE33, sometimes called ACTBP2 (Urquhart et al. 1993).

Not all alleles for an STR locus contain complete repeat units. Even simple repeats can contain non-consensus alleles that fall in between alleles with full repeat units. *Microvariants* are alleles that contain incomplete repeat units. Perhaps the most common example of a microvariant is the allele 9.3 at the TH01 locus, which contains nine tetranucleotide repeats and one incomplete repeat of three nucleotides because the seventh repeat is missing a single adenine out of the normal AATG repeat unit (Puers et al. 1993)—or a single thymine out of the TCAT repeat if the other strand is considered for nomenclature purposes.

STRs Used in Forensic DNA Typing

For human identification purposes it is important to have DNA markers that exhibit the highest possible variation or a number of less polymorphic markers that can be combined in order to obtain the ability to discriminate between samples. Forensic specimens are often challenging to PCR amplify because the DNA in the samples may be severely degraded (i.e., broken up into small pieces). Mixtures are prevalent as well in some forensic samples, such as those obtained from sexual assault cases containing biological material from both the perpetrator and victim.

The small size of STR alleles (≈100 bp to 400 bp) compared to minisatellite VNTR alleles (≈400 bp to 1000 bp) make the STR markers better candidates for use in forensic applications where degraded DNA is common. PCR amplification of degraded DNA samples can be better accomplished with smaller product sizes (see Chapter 10). These reduced-size STR amplicons are often referred to as *miniSTRs*. Allelic dropout of larger alleles in minisatellite markers caused by preferential amplification of the smaller allele is also a significant problem with minisatellites (Tully et al. 1993). Furthermore, single-base resolution of DNA fragments can be obtained more easily with sizes below 500 bp using high-resolution capillary electrophoresis (see Chapter 6). Thus, for both biology and technology reasons the smaller STRs are advantageous compared to the larger minisatellite VNTRs.

Among the various types of STR systems, tetranucleotide repeats have become more popular than di- or trinucleotides. Penta- and hexanucleotide repeats are less common in the human genome but are being examined by some laboratories (Bacher et al. 1999). A biological phenomenon known as "stutter" results when STR alleles are PCR amplified. *Stutter products* are amplicons that are typically one or more repeat units less in size than the true allele and arise during PCR because of strand slippage (Walsh et al. 1996).

Stutter product amounts vary depending on the STR locus and even the length of the allele within the locus but are usually less than 15% of the allele product quantity with tetranucleotide repeats. With di- and trinucleotides, the stutter percentage can be much greater (30% or more) making it difficult to interpret sample mixtures. In addition, the four-base spread in alleles with tetranucleotides makes closely spaced heterozygotes easier to resolve with size-based electrophoretic separations compared to alleles that could be two or three bases different in size with dinucleotide and trinucleotide markers, respectively.

GGAGGATGACTGTGTTCCCACTCTCAGTCCTGCCGAGGTGCCTGACAGCCCTG
CACCCAGGAGCTGGGGGGTCTAAGAGCTTGTAAAAAGTGTACAAGTGCCAGAT
GCTCGTTGTGCACAAATCTAAATGCAGAAAAGCACTGAAAGAAGAATCCCGAA
AACCACAGTTCCCATTTTTATATGGGAGCAAACAAAGCAGATCCCAAGCTCTT
CCTCTTCCCTAGATCAATACAGACAGACAGACAGGTG/gata/gata/gata/
gata/gata/gata/gata/gata/gata/gata/gata/TCATTGAAAGACA
AAACAGAGATGGATGATAGATACATGCTTACAGATGCACACACAAACGCTAAA
TGGTATAAAAATGGAATCACTCTGTAGGCTGTTTTACCACCTACTTTACTAAA
TTAATGAGTTATTGAGTATAATTTAATTTTATATACTAATTTGAAACTGTGTC
ATTAGGTTTTTAAGT

FIGURE 5.1 DNA sequence of a D16S539 allele containing 11 GATA repeats (shown in blue font). The STR repeat sequence is included in lowercase font with breaks between each repeat unit for emphasis. Underlined regions in green font indicate PowerPlex 16 primer binding sites (Krenke et al. 2002) with the shaded portion showing the PCR product, which is 288 bp (see Appendix 1). This top strand of the reference sequence is the reverse complement of GenBank entry AC024591 available from http://www.cstl.nist.gov/biotech/strbase/seq_ref.htm. The "T" shown in red font indicates a point mutation that resulted in a null allele with a previous primer (Nelson et al. 2002).

Thus, to summarize, the advantages of using tetranucleotide STR loci in forensic DNA typing over VNTR minisatellites or di- and trinucleotide repeat STRs include:

- A narrow allele size range that permits multiplexing;
- A narrow allele size range that reduces allelic dropout from preferential amplification of smaller alleles;
- The capability of generating small PCR product sizes that benefit the recovery of information from degraded DNA specimens; and
- Reduced stutter product formation compared to dinucleotide repeats that benefit the interpretation of sample mixtures.

A portion of the DNA sequence around the STR locus D16S539 is shown in Figure 5.1. This particular allele contains 11 GATA repeats and is 288 bp in length with the PCR primers highlighted (Krenke et al. 2002). PCR primers anneal to portions of the sequence on either side of the repeat region. In this manner, the variation that occurs between different alleles (e.g., 9 GATA repeats or 13 GATA repeats instead of the 11 GATA repeat units shown) will be captured by the overall size of the PCR product. Alleles differing by a single GATA repeat unit will be approximately 4 bp apart when separated by size using capillary electrophoresis (see Chapter 6). With these primers and 11 repeat units, there are 288 bp copied from the D16S539 locus. Thus, because allele 11 is 288 bp with the Promega primers, allele 10 will be 284 bp (4 bp smaller), allele 12 will be 292 bp (4 bp larger), and so forth. See Appendix 1 for a comprehensive listing of alleles and their sizes for each STR typing kit.

Using a different set of primers that are closer to the STR repeat region (Butler et al. 2003), the PCR product size for this same 11 GATA repeat allele is 105 bp in length (Figure 5.2).

In the past two decades, a number of tetranucleotide STRs have been explored for application to human identification. The types of STR markers that have been sought have

GGAGGATGACTGTGTTCCCACTCTCAGTCCTGCCGAGGTGCCTGACAGCCCTG
CACCCAGGAGCTGGGGGGTCTAAGAGCTTGTAAAAAGTGTACAAGTGCCAGAT
GCTCGTTGTGCACAAATCTAAATGCAGAAAAGCACTGAAAGAAGAATCCCGAA
AACCACAGTTCCCATTTTTATATGGGAGCAAACAAAGCAGATCCCAAGCTCTT
CCTCTTCCCTAGATCA**ATACAGACAGACAGACAGGTG**/gata/gata/gata/
gata/gata/gata/gata/gata/gata/gata/gata/TCATTGAAAGACA
AAAC**AGAGATGGATGATAGATACATGC**TTACAGATGCACACACAAACGCTAAA
TGGTATAAAAATGGAATCACTCTGTAGGCTGTTTTACCACCTACTTTACTAAA
TTAATGAGTTATTGAGTATAATTTAATTTTATATACTAATTTGAAACTGTGTC
ATTAGGTTTTTAAGT

FIGURE 5.2 DNA sequence of a D16S539 allele containing 11 GATA repeats. Underlined regions indicate miniSTR primer binding sites (Butler et al. 2003) with the shaded portion showing the PCR product, which is 105 bp, or a 183 bp size reduction over a PowerPlex 16 PCR product (see Figure 5.1).

included short STRs for typing degraded DNA materials (Coble & Butler 2005), STRs with low stuttering characteristics for analyzing mixtures (Bacher et al. 1999), and male-specific Y chromosome STRs for analyzing male-female mixtures from sexual crimes (Carracedo & Lareu 1998, Kayser et al. 2004). The selection criteria for candidate STR loci in human identification applications include the following characteristics (Gill et al. 1996, Carracedo and Lareu 1998):

- High discriminating power with observed heterozygosity 70%;
- Separate (or widely spaced) chromosomal locations to ensure that closely linked loci are not chosen;
- Robustness and reproducibility of results when multiplexed with other markers;
- Low stutter characteristics;
- Low mutation rate; and
- Predicted length of alleles that fall in the range of 90 bp to 500 bp with smaller sizes better suited for analysis of degraded DNA samples.

In order to take advantage of the product rule and be able to combine the genetic information across multiple loci, STR markers used in forensic DNA typing are typically chosen from separate chromosomes or are widely spaced on the same chromosome to avoid any problems with linkage between the markers.

CORE AND COMMON STR MARKERS

For DNA typing markers to be effective across a wide number of jurisdictions, a common set of standardized markers must be used. The STR loci that are commonly used today were initially characterized and developed either in the laboratory of Dr. Thomas Caskey at the Baylor College of Medicine (Edwards et al. 1991, Hammond et al. 1994) or at the Forensic Science Service (FSS) in England (Kimpton et al. 1993, Urquhart et al. 1994). The Promega Corporation (Madison, Wisconsin) initially commercialized many of the Caskey markers while Applied Biosystems (Foster City, California) incorporated the FSS STR loci and also developed some new markers.

Today both Applied Biosystems and the Promega Corporation have STR kits that address the needs of the DNA typing community and cover a common set of STR loci. The availability of STR kits that permit robust multiplex amplification of eight or more STR markers has truly revolutionized forensic DNA. Matching probabilities that exceed one in a billion are possible in a single amplification with 1 ng (or less) of DNA sample. Just as impressive is the fact that results can be obtained today in only a few hours compared to the weeks that restriction fragment length polymorphism (RFLP) methods took just a few years ago.

Early Developments

One of the first STR multiplexes to be developed was a quadruplex created by the Forensic Science Service that comprised the four loci TH01, FES/FPS, vWA, and F13A1 (Kimpton et al. 1994). This so-called "first-generation multiplex" had a matching probability of approximately 1 in 10,000. The FSS followed with a second-generation multiplex (SGM) made up of six polymorphic STRs and a gender identification marker (Kimpton et al. 1996, Sparkes et al. 1996). The six STRs in SGM were TH01, vWA, FGA, D8S1179, D18S51, and D21S11 and provided a matching probability of approximately 1 in 50 million. The gender identification marker amelogenin (Sullivan et al. 1993) will be described in more detail later in this chapter.

The first commercial STR kit capable of multiplex amplification for silver stain analysis became available from the Promega Corporation in 1994. This kit consisted of the STR loci CSF1PO, TPOX, and TH01 and is often referred to as the "CTT" triplex using the first letter in each locus. The CTT triplex only had a matching probability of ~1 in 500 but was still widely used in the United States in the mid-1990s as it was the first available STR multiplex kit and could be performed with a fairly low start-up cost. More information on early silver-stain and fluorescent STR multiplex assays is available in Chapter 3 of the *Fundamentals of Forensic DNA Typing* volume (Butler 2010).

The 13 CODIS STR Loci

In the United States, utilization of STRs initially lagged behind that of Europe, especially the efforts of the Forensic Science Service in the United Kingdom. However, beginning in 1996, the FBI Laboratory sponsored a community-wide forensic science effort to establish core STR loci for inclusion within the national DNA database known as CODIS (Combined DNA Index System).

Chapter 8 covers CODIS and DNA databases in more detail. This STR Project beginning in April 1996 and concluding in November 1997 involved 22 DNA typing laboratories and the evaluation of just 17 candidate STR loci: CSF1PO, F13A01, F13B, FES/FPS, FGA, LPL, TH01, TPOX, vWA, D3S1358, D5S818, D7S820, D8S1179, D13S317, D16S539, D18S51, and D21S11. Remember that the Human Genome Project was still ongoing and relatively few population studies had been conducted at that time.

At the STR Project meeting on November 13–14, 1997, 13 core STR loci were chosen to be the basis of the future CODIS national DNA database (Budowle et al. 1998). The 13 CODIS core loci are CSF1PO, FGA, TH01, TPOX, vWA, D3S1358, D5S818, D7S820, D8S1179, D13S317, D16S539, D18S51, and D21S11 (see Butler 2006). Table 5.1 lists the original references in the literature for

TABLE 5.1 Original Reference Describing Each of the 23 Common STR Loci and the Gender Identification Marker Amelogenin. Cooperative Human Linkage Center Information is Available via the Internet: http://www.chlc.org.

Locus Name	Reference
CSF1PO	Hammond, H.A., et al. (1994). Evaluation of 13 short tandem repeat loci for use in personal identification applications. *American Journal of Human Genetics, 55*, 175–189.
FGA	Mills, K.A., et al. (1992). Tetranucleotide repeat polymorphism at the human alpha fibrinogen locus (FGA). *Human Molecular Genetics, 1*, 779.
TH01	Polymeropoulos, M.H., et al. (1991). Tetranucleotide repeat polymorphism at the human tyrosine hydroxylase gene (TH). *Nucleic Acids Research, 19,* 3753.
SE33 (ACTBP2)	Polymeropoulos, M.H., et al. (1992). Tetranucleotide repeat polymorphism at the human beta-actin related pseudogene H-beta-Ac-psi-2 (ACTBP2). *Nucleic Acids Research, 20,* 1432.
TPOX	Anker, R., et al. (1992). Tetranucleotide repeat polymorphism at the human thyroid peroxidase (hTPO) locus. *Human Molecular Genetics, 1*, 137.
vWA	Kimpton, C.P., et al. (1992). A further tetranucleotide repeat polymorphism in the vWF gene. *Human Molecular Genetics, 1*, 287.
D1S1656	Cooperative Human Linkage Center GATA44E05.40831
D2S441	Cooperative Human Linkage Center GATA8F03.505
D2S1338	Cooperative Human Linkage Center GGAA3A09.31762
D3S1358	Li, H., et al. (1993). Three tetranucleotide polymorphisms for loci: D3S1352, D3S1358, D3S1359. *Human Molecular Genetics, 2*, 1327.
D5S818	Cooperative Human Linkage Center GATA3F03.512
D7S820	Cooperative Human Linkage Center GATA3F01.511
D8S1179	Cooperative Human Linkage Center GATA7G07.37564
D10S1248	Cooperative Human Linkage Center G00-366-431
D12S391	Cooperative Human Linkage Center GATA11H08.731
D13S317	Cooperative Human Linkage Center GATA7G10.415
D16S539	Cooperative Human Linkage Center GATA11C06.715
D18S51	Staub, R.E., et al. (1993). A microsatellite genetic linkage map of human chromosome 18. *Genomics, 15*, 48–56.
D19S433	Cooperative Human Linkage Center GGAA2A03.135
D21S11	Sharma, V., & Litt, M. (1992). Tetranucleotide repeat polymorphism at the D21S11 locus. *Human Molecular Genetics, 1*, 67.
D22S1045	Cooperative Human Linkage Center ATA37D06.40596
Penta D	Bacher, J., & Schumm, J.W. (1998). Development of highly polymorphic pentanucleotide tandem repeat loci with low stutter. *Profiles in DNA, 2(2),* 3–6.
Penta E	Bacher, J., & Schumm, J.W. (1998). Development of highly polymorphic pentanucleotide tandem repeat loci with low stutter. *Profiles in DNA, 2(2),* 3–6.
Amelogenin	Sullivan, K.M., et al. (1993). A rapid and quantitative DNA sex test: fluorescence-based PCR analysis of X-Y homologous gene amelogenin. *BioTechniques, 15*, 637–641.

these 13 STRs. When all 13 CODIS core loci are tested, the average random match probability is rarer than one in a trillion among unrelated individuals (Chakraborty et al. 1999).

Of the original 13 CODIS STR loci, the three most polymorphic markers are FGA, D18S51, and D21S11. TPOX, CSF1PO, and TH01 typically exhibit the least amount of variation between individuals. A summary of information on the 13 CODIS STRs (and 10 other commonly used loci) is contained in Table 5.2, which describes the chromosomal location and physical location on the human genome reference sequence, the repeat motif, the allele range, and the GenBank accession number where the DNA sequence for a reference allele may be found. The chromosomal locations for these STRs have been updated on the completed human genome reference sequence. Detailed allele sequence information and PCR product sizes with commercially available STR kits may be found in Appendix 1.

For those who wonder why poor-performing loci such as CSF1PO and TPOX were selected, we can go back to a presentation made at the October 1995 Promega meeting. The state of Alabama was an early adopter of STR typing technology, which at that time only had silver-stain triplexes available. By October 1995, Alabama had tested 13,610 offenders with CSF1PO, TPOX, and TH01 (Hicks 1995). At that point in time, this would probably have been the largest set of DNA samples containing STR data in the United States.

The Deputy Director of the Alabama Department of Forensic Sciences in 1995 was the former FBI Laboratory director, John W. Hicks. Director Hicks actually writes that these offender samples had been examined "under the assumption that STR systems will soon be embraced by CODIS *which will include the TH01, TPOX and CSF1PO loci*" (Hicks 1995, emphasis added). This statement was recorded more than two years before the original 13 CODIS STR loci were officially selected. While these loci are perhaps not optimal for human identity testing, we may be stuck with them now that almost 10 million profiles have been run in the United States.

Using the previously described classification scheme for categorizing STR repeat motifs (Urquhart et al. 1994), the 13 CODIS core STR loci may be divided up into four categories:

1. Simple repeats consisting of one repeating sequence: TPOX, CSF1PO, D5S818, D13S317, D16S539;
2. Simple repeats with non-consensus alleles (e.g., 9.3): TH01, D18S51, D7S820;
3. Compound repeats with non-consensus alleles: vWA, FGA, D3S1358, D8S1179; and
4. Complex repeats: D21S11.

The European Standard Set

European forensic DNA laboratories utilize many of the same STR loci as used in the United States. Originally building on the initial FSS work, a European Standard Set (ESS) of STR loci were selected in 1999 (Schneider 2009). The original ESS included FGA, TH01, vWA, D3S1358. D8S1179, D18S51, and D21S11. These seven STR loci are the six SGM markers plus D3S1358. When D16S539, which is part of the CODIS core loci and present in many STR typing kits, is analyzed, there are eight STR loci that overlap between European and U.S. DNA tests.

The Interpol international DNA database requires testing at least the ESS so that seven STR loci will be in common when comparisons are made. Thus, the ESS are the same as the

TABLE 5.2 Information on Commonly Used Autosomal STR Loci. The 13 CODIS Core Loci Are Highlighted in Bold Font.

Locus (UniSTS)*	Chromosomal Location	Physical Position GRCh37 Assembly	GenBank Accession (allele repeat #)	Category and Repeat Motif	Allele Range (Appendix 1)
D1S1656 (58809)	1q42	Chr 1 230.905 Mb	G07820 (15.3)	compound TAGA	8 to 20.3
TPOX (240638)	2p25.3 thyroid peroxidase, 10th intron	Chr 2 1.493 Mb	M68651 (11)	simple AATG	4 to 16
D2S441 (71306)	2p14	Chr 2 68.239 Mb	AC079112 (12)	compound TCTA/TCAA	8 to 17
D2S1338 (30509)	2q35	Chr 2 218.879 Mb	AC010136 (23)	compound TGCC/TTCC	10 to 31
D3S1358 (148226)	3p21.31	Chr 3 45.582 Mb	AC099539 (16)	compound TCTA/TCTG	6 to 26
FGA (240635)	4q31.3 alpha fibrinogen, 3rd intron	Chr 4 155.509 Mb	M64982 (21)	compound CTTT/TTCC	12.2 to 51.2
D5S818 (54700)	5q23.2	Chr 5 123.111 Mb	AC008512 (11)	simple AGAT	4 to 29
CSF1PO (156169)	5q33.1 c-fms proto-oncogene, 6th intron	Chr 5 149.455 Mb	X14720 (12)	simple AGAT	5 to 17
SE33 (ACTBP2) (none reported)	6q14 beta-actin related pseudogene	Chr 6 88.987 Mb	V00481 (26.2)	complex AAAG	3 to 49
D7S820 (74895)	7q21.11	Chr 7 83.789 Mb	AC004848 (13)	simple GATA	5 to 16
D8S1179 (83408)	8q24.13	Chr 8 125.907 Mb	AF216671 (13)	compound TCTA/TCTG	6 to 20
D10S1248 (51457)	10q26.3	Chr 10 131.093 Mb	AL391869 (13)	simple GGAA	7 to 19
TH01 (240639)	11p15.5 tyrosine hydroxylase, 1st intron	Chr 11 2.192 Mb	D00269 (9)	simple TCAT	3 to 14
vWA (240640)	12p13.31 von Willebrand Factor, 40th intron	Chr 12 6.093 Mb	M25858 (18)	compound TCTA/TCTG	10 to 25
D12S391 (2703)	12p13.2	Chr 12 12.450 Mb	G08921(20)	compound AGAT/AGAC	13 to 27.2
D13S317 (7734)	13q31.1	Chr 13 82.692 Mb	AL353628 (11)	simple TATC	5 to 17
Penta E (none reported)	15q26.2	Chr 15 97.374 Mb	AC027004 (5)	simple AAAGA	5 to 32
D16S539 (45590)	16q24.1	Chr 16 86.386 Mb	AC024591 (11)	simple GATA	4 to 17
D18S51 (44409)	18q21.33	Chr 18 60.949 Mb	AP001534 (18)	simple AGAA	5.3 to 40
D19S433 (33588)	19q12	Chr 19 30.416 Mb	AC008507 (14)	compound AAGG/TAGG	5.2 to 20
D21S11 (240642)	21q21.1	Chr 21 20.554 Mb	AP000433 (29.1)	complex TCTA/TCTG	12 to 43.2
Penta D (none reported)	21q22.3	Chr 21 45.056 Mb	AP001752 (13)	simple AAAGA	1.1 to 19
D22S1045 (49680)	22q12.3	Chr 22 37.536 Mb	AL022314 (17)	simple ATT	7 to 20

Adapted from Butler (2006). Physical positions are from Thanakiatkrai & Welch (2010).

**UniSTS is a comprehensive database of sequence tagged sites (STSs) available on the NCBI website: http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?db=unists.*

Interpol Standard Set of Loci (ISSOL). However, as described in Chapter 8, when DNA databases grow in numbers, more loci are required to avoid adventitious matches. In Germany, the highly polymorphic locus SE33 was adopted as part of their national DNA database.

At the European Network of Forensic Science Institutes (ENFSI) meeting in April 2005, several new European STR loci were recommended for inclusion in future European STR typing kits. These include D2S441, D10S1248, D22S1045, D1S1656, and D12S391 (Gill et al. 2006a, Gill et al. 2006b). In April 2009, ENFSI voted to extend the ESS to 12 loci—the original ESS seven plus the five recommended loci. The extended ESS loci were formally adopted by the European Union in November 2009 to enable DNA data exchange across Europe.

The Expanded U.S. Core Loci

With the U.S. national DNA database surpassing nine million profiles by the end of 2010, the FBI Laboratory began exploring possible loci to expand the U.S. core set. In order to permit more international comparisons when needed, the expanded ESS loci are being considered to extend the U.S. core beyond the original 13 STRs to 18 or more STRs. The STR loci under consideration are part of the 23 markers described in Table 5.2 and utilized in various commercially available kits. At the time this book was completed in early 2011, the final set of expanded U.S. core loci had not yet been selected. More information on the U.S. core loci can be found at http://www.fbi.gov/about-us/lab/codis.

COMMERCIAL STR KITS

A typical STR typing kit consists of the following five components: (1) a **PCR primer mixture** containing oligonucleotides designed to amplify a set of STR loci (one of each pair of primers is labeled with a fluorescent dye), (2) a **PCR buffer** containing deoxynucleotide triphosphates, $MgCl_2$, and other reagents necessary to perform PCR, (3) a **DNA polymerase**, which is sometimes premixed with the PCR buffer, (4) an **allelic ladder** with common alleles for the STR loci being amplified to enable calibration of allele repeat size, and (5) a **positive control DNA** sample to verify that the kit reagents are working properly.

A number of kits are available for single or multiplex PCR amplification of STR markers used in DNA typing. Two primary vendors for STR kits used by the forensic DNA community exist: the Promega Corporation located in Madison, Wisconsin, and Applied Biosystems located in Foster City, California. These companies have expended a great deal of effort over the past 15 years to bring STR markers to forensic scientists in kit form. More recently in Europe, companies such as Serac (Bad Homburg, Germany) and Biotype (Dresden, Germany) have begun offering commercial STR kits, but due to patent and licensing issues some of these kits have limited distribution. For example, QIAGEN (Hilden, Germany) sells their Investigator Human Identification PCR kits, which were originally developed by Biotype, in Europe but not in the United States. A list of commercially available STR multiplexes and when they were released as products is shown in Table 5.3.

The adoption of the 13 core loci for CODIS in the United States has led to development of STR multiplexes that cover these markers. At the turn of the century, two PCR reactions were required to obtain information from all 13 STRs: either PowerPlex 1.1 and PowerPlex

TABLE 5.3 Commercially Available Autosomal STR Multiplexes (Fluorescently-Labeled). Kits with Improved Buffers That Overcome PCR Inhibitors Are Listed in Bold Font.

Name	Source	Release Date
AmpFlSTR Blue *(no longer available)*	Applied Biosystems	Oct 1996
AmpFlSTR Green I *(no longer available)*	Applied Biosystems	Jan 1997
CTTv	Promega	Jan 1997
FFFL	Promega	Jan 1997
GammaSTR	Promega	Jan 1997
PowerPlex 1.1, PowerPlex 1.2	Promega	Jan 1997, Sept 1998
AmpFlSTR Profiler	Applied Biosystems	May 1997
AmpFlSTR Profiler Plus	Applied Biosystems	Dec 1997
AmpFlSTR COfiler	Applied Biosystems	May 1998
AmpFlSTR SGM Plus	Applied Biosystems	Feb 1999
PowerPlex 2.1 (for Hitachi FMBIO users)	Promega	June 1999
PowerPlex 16	Promega	May 2000
PowerPlex 16 BIO (for Hitachi FMBIO users)	Promega	May 2001
AmpFlSTR Identifiler	Applied Biosystems	July 2001
AmpFlSTR Profiler Plus ID (extra unlabeled D8-R primer)	Applied Biosystems	Sept 2001
PowerPlex ES	Promega	Mar 2002
AmpFlSTR SEfiler *(no longer available)*	Applied Biosystems	Sept 2002
AmpFlSTR MiniFiler	Applied Biosystems	Mar 2007
AmpFlSTR SEfiler Plus *(improved buffer)*	Applied Biosystems	Nov 2007
AmpFlSTR Sinofiler *(labeling & documentation only available in Chinese)*	Applied Biosystems	Mar 2008
PowerPlex 16 HS *(same primers, improved reagents)*	Promega	Mar 2009
PowerPlex ESX 16 & ESX 17	Promega	Sept 2009
PowerPlex ESI 16 & ESI 17	Promega	Sept 2009
AmpFlSTR Identifiler Direct *(same primers, improved reagents)*	Applied Biosystems	Nov 2009
AmpFlSTR Identifiler Plus *(same primers, improved reagents)*	Applied Biosystems	Jan 2010
AmpFlSTR NGM	Applied Biosystems	Jan 2010
Investigator ESSplex	QIAGEN	April 2010
Investigator Decaplex SE	QIAGEN	April 2010
Investigator Triplex AFS QS	QIAGEN	April 2010
Investigator Triplex DSF	QIAGEN	April 2010

(*Continued*)

TABLE 5.3 Commercially Available Autosomal STR Multiplexes (Fluorescently-Labeled). Kits with Improved Buffers That Overcome PCR Inhibitors Are Listed in Bold Font. (*Continued*)

Name	Source	Release Date
Investigator IDplex	QIAGEN	Aug 2010
Investigator HDplex	QIAGEN	Sept 2010
Investigator Hexaplex ESS	QIAGEN	Sept 2010
Investigator Nonaplex ESS	QIAGEN	Sept 2010
Investigator ESSplex SE	QIAGEN	Oct 2010
AmpFlSTR NGM SElect	Applied Biosystems	Dec 2010
PowerPlex 18D	Promega	Feb 2011

2.1 or Profiler Plus and COfiler. As an internal check to reduce the possibility of mixing up samples, both manufacturers included overlapping loci in their kits that should produce concordant data between samples amplified from the same biological material. The Profiler Plus and COfiler kits have the loci D3S1358 and D7S820 (and the sex-typing marker amelogenin) in common while the PowerPlex 1.1 and PowerPlex 2.1 Systems have the loci TH01, TPOX, and vWA in common.

Since 2000, both Promega and Applied Biosystems have marketed multiplex PCR reactions that permit co-amplification of all 13 STRs in a single reaction along with the amelogenin sex-typing marker and two additional STR loci.

For the first decade or so of STR use, two primary methods were applied to separate and detect fluorescently labeled STR alleles: gel electrophoresis and capillary electrophoresis (see Butler 2010, *Fundamentals*, Chapter 9). Several of the PowerPlex kits were balanced to work with the Hitachi FMBIO scanner following polyacrylamide gel electrophoresis while PowerPlex 16, Identifiler, Profiler Plus, and COfiler reactions are typically analyzed on an ABI Prism 310 or 3100 Genetic Analyzer capillary electrophoresis system (see Chapter 6). Today capillary electrophoresis is exclusively used for routine STR testing.

Commercial manufacturers of STR kits have spent a great deal of research effort defining which markers would be included in each kit as well as verifying if primer pairs are compatible and will work well in combination with each other during multiplex PCR conditions (Wallin et al. 2002, Krenke et al. 2002). Promega has published and patented their PCR primer sequences (Masibay et al. 2000, Krenke et al. 2002) whereas Applied Biosystems have kept their primer sequences proprietary although some information has been revealed regarding the use of degenerate primers (Leibelt et al. 2003). The issue over failure to disclose kit primer sequences impacted several court cases early on in the legal acceptance of STR technology but appears to have been resolved now (see Butler 2010, *Fundamentals* D.N.A. Box 8.2).

Most laboratories do not have the time or resources to design primers, optimize PCR multiplexes, and monitor the quality control of primer synthesis. The convenience of using ready-made kits is also augmented by the fact that widely used primer sets and conditions allow improved opportunities for sharing data between laboratories without fear of possible

null alleles. Available STR multiplex sets vary based on which STR loci are included, the fluorescent dye combinations, the DNA strand that is labeled, allelic ladders present in kits, and most importantly, the primer sequences utilized for PCR amplification. It is important to keep in mind that commercially available kits quickly dictate which STRs will be used by the vast majority of forensic laboratories.

Allelic Ladders

An allelic ladder is an artificial mixture of the common alleles present in the human population for a particular STR marker (Sajantila et al. 1992). They are generated with the same primers as tested samples and thus provide a reference DNA size for each allele included in the ladder. Allelic ladders have been shown to be important for accurate genotype determinations (Smith 1995, Kline et al. 1997). These allelic ladders serve as a standard like a ruler for each STR locus. They are necessary to adjust for different sizing measurements obtained from different instruments and conditions used by various laboratories.

Allelic ladders are constructed by combining genomic DNA or locus-specific PCR products from multiple individuals in a population, which possess alleles that are representative of the variation for the particular STR marker (Sajantila et al. 1992, Baechtel et al. 1993). The samples are then co-amplified to produce an artificial sample containing the common alleles for the STR marker. Allele quantities are balanced by adjusting the input amount of each component so that the alleles are fairly equally represented in the ladder. For example, to produce a ladder containing five alleles with 6, 7, 8, 9, and 10 repeats, individual samples with genotypes of (6,8), (7,10), and (9,9) could be combined. Alternatively, the combination of genotypes could be (6,9), (7,8), and (10,10) or (6,6), (7,7), (8,8), (9,9), and (10,10).

Additional quantities of the same allelic ladder (second- and third-generation ladders) may be produced by simply diluting the original ladder 1/1000 to 1/1000000 parts with deionized water and then re-amplifying it using the same PCR primers (Baechtel et al. 1993). It is imperative that allelic ladders be generated with the same PCR primers as used to amplify unknown samples so that the allele "rung" on the ladder will accurately line up with that of the repeat number of the unknown sample when the unknown is compared to the ladder.

Commercial manufacturers provide allelic ladders in their STR typing kits so that individual laboratories do not have to produce their own allelic ladders. Manufacturing processes have improved so that well-balanced allelic ladders can be consistently provided to the forensic DNA community. Kits from Applied Biosystems, Promega Corporation, or QIAGEN for comparable STR markers often contain different alleles in their allelic ladders. For example, the Promega PowerPlex ESI 17 allelic ladder for D16S539 contains consecutive alleles ranging from 4 to 16 repeats while the QIAGEN ESSplex D16S539 allelic ladder contains 8 to 16 repeats (Figure 5.3). Presumably QIAGEN did not have access to D16S539 samples with 4, 5, 6, or 7 repeats when their allelic ladders were developed.

AmpFlSTR Identifiler Kit Innovations

Applied Biosystems introduced two new technologies with their AmpFlSTR Identifiler kit when it was released in 2001 (Applied Biosystems 2001, Collins et al. 2004). The first, and most obvious, involved the use of five-dye detection systems where four different dyes (6FAM,

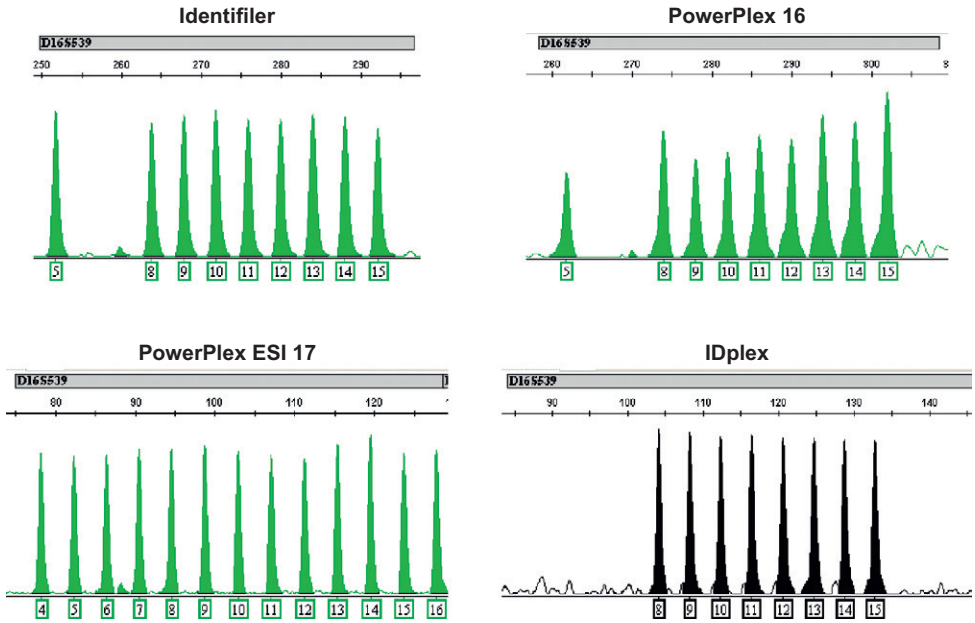


FIGURE 5.3 D16S539 allelic ladders for Identifiler (ABI), PowerPlex 16 (Promega), PowerPlex ESI 17 (Promega), and IDplex (QIAGEN). Figure courtesy of Becky Hill, NIST.

VIC, NED, and PET) are used to label the PCR products rather than the traditional three dyes (5FAM, JOE, NED or FL, JOE, TMR) as used with the previous AmpFlSTR or PowerPlex kits. One dye detection channel is always used for an internal size standard to correlate electrophoretic mobilities to an apparent PCR product size. Thus, the fifth dye (LIZ) in 5-dye detection and the fourth dye (ROX or CXR) in 4-dye detection are used for labeling the internal size standard. The extra dye channel for labeling PCR products enables smaller PCR products to be generated and placed in a separate dye channel rather than extending the size range for amplicons within the three previously available dye channels.

The second technology introduced with the Identifiler kit involves mobility-modifying non-nucleotide linkers (Applied Biosystems 2001). The mobility modifier is composed of hexaethyleneoxide (HEO) that imparts a shift of approximately 2.5 nucleotides with each additional HEO unit (Grossman et al. 1994). This non-nucleotide linker is synthesized into the 5′-end of the PCR primer so that when the PCR product is created it contains these extra molecules on one end (see Figure 4.4). By incorporating non-nucleotide linkers, mobilities for amplified alleles from one member of a pair of closely spaced STR loci can be shifted relative to the other. Thus, overlapping size ranges can be prevented.

The primary reason for introducing mobility modifiers is to permit continued use of the same PCR primers for amplifying STR loci and still have optimal inter-locus spacing within the various color channels. For example, if the loci D7S820 and CSF1PO, which are labeled with two different fluorophores in the COfiler kit and therefore do not interfere with one another, were labeled with the same colored fluorescent label (e.g., 6FAM) as they are in

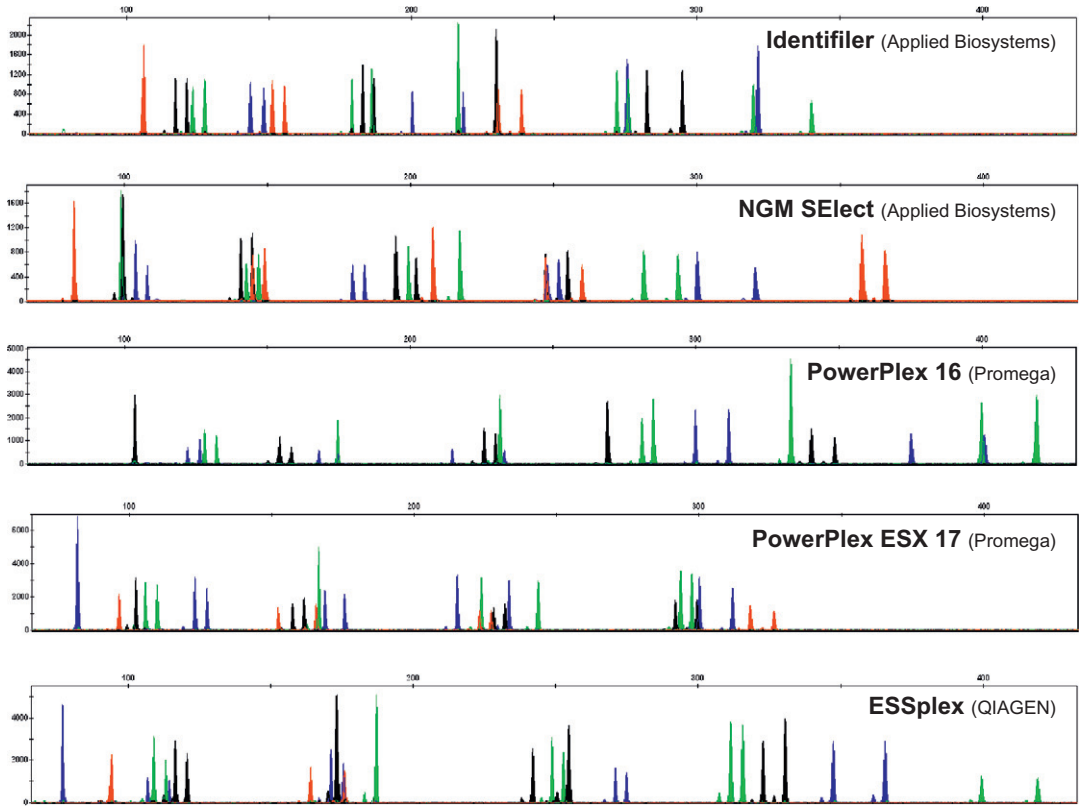


FIGURE 5.4 The same female DNA sample amplified with five different STR typing kits. Internal size standards are not shown. The PowerPlex 16 kit uses three fluorescent dye colors to label its PCR products while all of the other kits utilize four dyes. Figure courtesy of Becky Hill, NIST.

the Identifiler STR kit, the allelic ladder products would have overlapped by approximately 13bp. To prevent this overlap in allele size ranges, either PCR primer binding sites must be altered to change the overall size of the PCR product or mobility modifiers can be introduced to shift the apparent molecular weight of the larger PCR product to an even larger size. In the case of the Identifiler kit, the locus CSF1PO was shifted by approximately 25bp—through the addition of 10 HEO non-nucleotide linkers to the 5′-end of the labeled PCR primer. Non-nucleotide linkers are also present on four other loci in the Identifiler kit: D2S1338, D13S317, D16S539, and TPOX.

Promega has changed primer sequences for a few of the loci between PowerPlex versions (see Masibay et al. 2000, Butler et al. 2001, Krenke et al. 2002). For example, between the PowerPlex 1.1 and PowerPlex 16 kits, the CSF1PO primer positions were drastically altered in order to achieve a 30bp shift in PCR product size between the two kits. This primer change and subsequent PCR product shift was instituted so that CSF1PO and D16S539 loci could be labeled with the same dye in the PowerPlex 16 kit. Note that if the original CSF1PO primers had been kept, there would have been a 13bp overlap between D16S539 allele 15 (304bp) and

CSF1PO allele 6 (291 bp) making these systems incompatible in the same dye color without altering the PCR product size (i.e., primer positions) for one of them.

Different primer positions have the potential to lead to allele dropout if a primer binding site mutation impacts one of the primer pairs. Hence concordance studies are needed between various STR kits to assess the level of potential allele dropout (Budowle et al. 2001, Hill et al. 2010). Applied Biosystems has maintained the same primers over time and through their various AmpFlSTR kits (Holt et al. 2002) by introducing five-dye chemistry and mobility modifiers for products that would normally overlap with one another. With very few exceptions (e.g., MiniFiler primers and SE33 in NGM SElect), Applied Biosystems has maintained its primer sequences between kits helping to ensure consistency of results across kits (see Hill et al. 2010).

Next-Generation STR Kits

Although the contents of newer and what might be termed "next-generation" STR typing kits have not been fully disclosed by their manufacturers, they perform like turbo-charged engines with increased capabilities for improved performance particularly in the presence of PCR inhibitors. As discussed in Chapter 4, these improvements may include the presence of new DNA polymerases, additives to the PCR reaction, substitution of high-stability nucleic acid analogs along the primers to improve their annealing stability (Mulero & Hennessy 2009, Ballantyne et al. 2010), and energy-transfer dye-labeled primers to improve PCR product sensitivity (Yeung et al. 2008).

The next-generation kits containing improved buffer formulations are listed in bold font in Table 5.3. They include (in order of release date): MiniFiler, SEfiler Plus, PowerPlex 16 HS, PowerPlex ESX 16/17, PowerPlex ESI 16/17, Identifiler Direct, Identifiler Plus, NGM, and NGM SElect. These next-generation kits have shown the capability of producing STR profiles in the presence of PCR inhibitors such as heme, humic acid, and tanic acid. In addition, because of their increased tolerance of PCR inhibitors these kits have enabled direct PCR from bloodstains without DNA extraction and purification (see Chapter 4). New STR loci have also been introduced with these new kits. Figure 5.4 compares electropherograms for the same source DNA for five of these systems.

In 2010, QIAGEN began selling STR typing kits in Europe. However, due to patent restrictions (e.g., in the U.S. and Japan), these kits are not sold worldwide. In the U.S., Identifiler and PowerPlex 16 are most commonly used while in Europe SGM Plus and SEfiler Plus were used prior to the more recently released NGM and PowerPlex ESI/ESX kits (Table 5.4).

Figure 5.5 contains a summary of STR loci (including dye label and relative size position) in 25 STR kits available from Applied Biosystems, Promega Corporation, and QIAGEN. The next section describes each of the 23 commonly used STR loci in more detail.

INFORMATION ON 23 CORE AND COMMONLY USED STR LOCI

Each of the 23 commonly used STR loci has unique characteristics, either in terms of the number of alleles present, the type of repeat sequence, or the kinds of microvariants that

TABLE 5.4 Comparison of STR Loci Present in Kits Used in the U.S. and Europe. The Eight Shaded Loci Overlap.

U.S.			Europe				
PP16	**Identifiler**	**MiniFiler**	**ESX/ESI17**	**NGM (SElect)**	**SEfiler Plus**	**SGM Plus**	**ESSplex**
TPOX	TPOX						
CSF1PO	CSF1PO	CSF1PO					
D5S818	D5S818						
D7S820	D7S820	D7S820					
D13S317	D13S317	D13S317					
FGA	FGA	FGA	FGA	FGA	FGA	FGA	FGA
vWA	vWA		vWA	vWA	vWA	vWA	vWA
D3S1358	D3S1358		D3S1358	D3S1358	D3S1358	D3S1358	D3S1358
D8S1179	D8S1179		D8S1179	D8S1179	D8S1179	D8S1179	D8S1179
D18S51	D18S51	D18S51	D18S51	D18S51	D18S51	D18S51	D18S51
D21S11	D21S11	D21S11	D21S11	D21S11	D21S11	D21S11	D21S11
TH01	TH01		TH01	TH01	TH01	TH01	TH01
D16S539	D16S539	D16S539	D16S539	D16S539	D16S539	D16S539	D16S539
	D2S1338	D2S1338	D2S1338	D2S1338	D2S1338	D2S1338	D2S1338
	D19S433		D19S433	D19S433	D19S433	D19S433	D19S433
			D12S391	D12S391			D12S391
			D1S1656	D1S1656			D1S1656
			D2S441	D2S441			D2S441
			D10S1248	D10S1248			D10S1248
			D22S1045	D22S1045			D22S1045
			SE33	(SE33)	SE33		
Penta D							
Penta E							

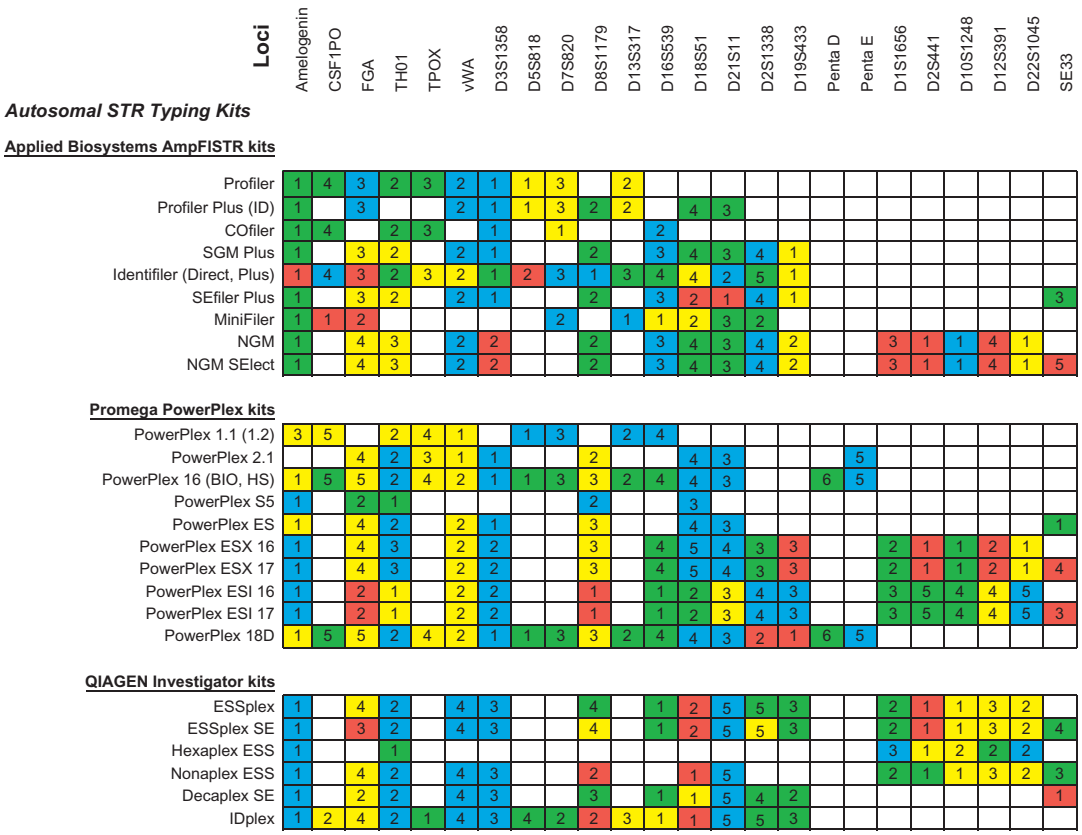


FIGURE 5.5 Commercially available STR typing kits with dye labels and size positions for loci present in the kit.

have been observed. This section reviews some of the basic details on each of the core and commonly used STR loci that are present in commercial STR kits.

Appendix 1 contains a detailed summary of the alleles that have been reported as of late 2010 for the 13 CODIS and other commonly used STR loci along with their expected sizes using various kits that are available from Promega or Applied Biosystems. The size difference in the PCR products produced by the different STR kits is important because a large difference is more likely to lead to null alleles when comparing results between two kits.

The 13 CODIS STR Loci

CSF1PO is a simple tetranucleotide repeat found in the sixth intron of the c-fms proto-oncogene for the CSF-1 receptor on the long arm of chromosome 5. Common alleles contain an AGAT core repeat and range in size from 6 to 15 repeats. An allele 16 has also been reported (Margolis-Nunno et al. 2001) as have several x.1 and x.3 variant alleles (see Appendix 1). PCR products from Promega's PowerPlex 1.1 STR kit are 11 bp larger than

those generated with Applied Biosystems kits for equivalent alleles. Since PowerPlex 16 adds 30 bp to the size of CSF1PO relative to PowerPlex 1.1, the PowerPlex 16 PCR products for CSF1PO are 41 bp larger than those generated with AmpFlSTR kits. Mobility modifiers are included with CSF1PO in the Identifiler kit to increase the apparent PCR product size by around 25 bp (see Applied Biosystems 2001).

FGA is a compound tetranucleotide repeat found in the third intron of the human alpha fibrinogen locus on the long arm of chromosome 4. FGA has also been referred to in the literature as FIBRA or HUMFIBRA. The locus contains a CTTT repeat flanked on either side by degenerate repeats. The spread in allele sizes is larger for FGA than for any of the other core STR loci. Reported alleles range in size from 12.2 repeats to 51.2 repeats, spanning over 35 repeats! A 2 bp deletion, from the loss of a CT, in the region just prior to the core repeat motif is responsible for the x.2 microvariant alleles that are very prevalent in this STR system. PCR products from Promega's PowerPlex 2.1 and 16 STR kits are 112 bp larger than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. This size difference between these two primer sets is the largest of any of the 13 core loci. So far a total of 95 different FGA alleles have been reported (see Appendix 1) making it one of the most polymorphic loci used in human identity testing.

TH01 is a simple tetranucleotide repeat found in intron 1 of the tyrosine hydroxylase gene on the short arm of chromosome 11. The locus name arises from the initials for tyrosine hydroxylase and intron 1 (i.e., 01). The locus is sometimes incorrectly referred to as "THO1" with a letter "O" instead of a "zero." In the literature, TH01 has also been referred to as TC11 and HUMTH01.

TH01 has a simple tetranucleotide sequence with a repeat motif of TCAT on the upper strand in the GenBank reference sequence. The repeat motif is commonly referenced as AATG, which is correct for the complementary (bottom) strand to the GenBank reference sequence. A common microvariant allele that exists in Caucasians contains a single base deletion from allele 10 and is designated allele 9.3. Other x.3 alleles have been reported such as 8.3, 10.3, and 13.3 (Griffiths et al. 1998). TH01 has probably been the most studied of the 13 core loci with over 1000 population studies reported in the literature using this DNA marker. PCR products from Promega's PowerPlex 1.1 STR kit are 11 bp larger than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. PowerPlex 2.1 STR kits produce amplicons that are 19 bp smaller than PowerPlex 1.1. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for TH01 (Butler et al. 2001, Krenke et al. 2002).

TPOX is a simple tetranucleotide repeat found in intron 10 of the human thyroid peroxidase gene near the very end of the short arm of chromosome 2. TPOX has also been referred to in the literature as hTPO. This STR locus possesses a simple AATG repeat and is the least polymorphic of the 13 core loci. PCR products from Promega's PowerPlex 1.1 STR kit are 7 bp larger than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. PowerPlex 2.1 STR kits produce amplicons that are 38 bp larger in size relative to PowerPlex 1.1. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for TPOX (Masibay et al. 2000, Krenke et al. 2002). Tri-allelic (three banded) patterns are more prevalent in TPOX than in any other forensic STR marker.

vWA is a compound tetranucleotide repeat found in intron 40 of the von Willebrand Factor gene on the short arm of chromosome 12. vWA has also been referred to in the literature as vWF and vWA. It possesses a TCTA repeat interspersed with a TCTG repeat. The vWA

marker targeted by STR multiplex kits is only one of three repeats present in that region of the von Willebrand Factor. The other two have not been found to be as polymorphic (Kimpton et al. 1992). PCR products from Promega's PowerPlex 1.1 STR kit are 29 bp smaller than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. The PowerPlex 1.1 and PowerPlex 2.1 STR kits overlap at three STRs including vWA. Both kits produce amplicons that are equivalent in size for vWA alleles. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for vWA (Butler et al. 2001, Krenke et al. 2002).

D3S1358 is a compound tetranucleotide repeat found on the short arm of chromosome 3. This locus possesses both AGAT and AGAC repeat units (Mornhinweg et al. 1998). The D3 marker is common to Applied Biosystems' AmpFlSTR multiplexes Blue, Profiler, Profiler Plus, COfiler, SGM Plus, SEfiler, Identifiler, and NGM. PCR products from Promega's PowerPlex 2.1 STR kit are 2 bp larger than those generated with Applied Biosystems kits for equivalent alleles. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for D3S1358 (Butler et al. 2001, Krenke et al. 2002).

D5S818 is a simple tetranucleotide repeat found on the long arm of chromosome 5. The locus possesses AGAT repeat units with alleles ranging in size from 7 to 16 repeats. In both Promega and Applied Biosystems STR kits, D5S818 is one of the smaller sized loci and as such should appear more often than some of the other loci in degraded DNA samples. Only a few rare microvariants have been reported at this STR marker. PCR products from Promega's PowerPlex 1.1 STR kit are 15 bp smaller than those generated with Applied Biosystems kits for equivalent alleles and PowerPlex 16 retains the original PowerPlex 1.1 primers (Masibay et al. 2000, Krenke et al. 2002).

D7S820 is a simple tetranucleotide repeat found on the long arm of chromosome 7. The locus possesses primarily a GATA repeat. However, a number of new D7 microvariant alleles have been reported recently (see Appendix 1). These x.1 and x.3 alleles likely result due to a variation in the number of T nucleotides found in a poly(T) stretch that occurs 13 bases downstream of the core GATA repeat. Sequencing has revealed that "on-ladder" alleles contain nine tandem T's while x.3 alleles contain eight T's and x.1 alleles contain 10 T's (Egyed et al. 2000). PCR products from Promega's PowerPlex 1.1 STR kit are 42 bp smaller than those generated with Applied Biosystems kits for equivalent alleles.

D8S1179 is a compound tetranucleotide repeat found on chromosome 8. In early publications by the Forensic Science Service, D8S1179 is listed as D6S502 because of a labeling error in the Cooperative Human Linkage Center database from which this STR was chosen (Oldroyd et al. 1995, Barber & Parkin 1996). The locus consists primarily of alleles containing TCTA although a TCTG repeat unit enters the motif for all alleles larger than 13 repeats, usually at the second or third position from the 5′-end of the repeat region (Barber & Parkin 1996). PCR products from Promega's PowerPlex 2.1 and PowerPlex 16 STR kits are 80 bp larger than those generated with Applied Biosystems kits for equivalent alleles. AmpFlSTR Identifiler and Profiler Plus ID kits possess an extra, unlabeled D8S1179 reverse primer to prevent allele drop-out in a small portion of Asian populations due to a mutation in the middle of the primer-binding site (Leibelt et al. 2003).

D13S317 is a simple TATC tetranucleotide repeat found on the long arm of chromosome 13. Common alleles contain between 7 to 15 repeat units although alleles 5, 6, and 16 have been reported (see Appendix 1). PCR products from Promega's PowerPlex 1.1 STR kit are 36 bp smaller than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles.

A 4 bp deletion has been reported 24 bases downstream from the core TATC repeat that can impact allele calls with different primer sets (Butler et al. 2003, Drábek et al. 2004). PowerPlex 16 primers (Krenke et al. 2002), while generating the same size amplicons as the original PowerPlex 1.1 primers (Masibay et al. 2000), have been shifted to avoid this 4 bp deletion that is present in some African-American samples (Krenke et al. 2002).

D16S539 is one of the core U.S. loci and present in most commercial STR kits. It is a simple tetranucleotide repeat found on the long arm of chromosome 16. Nine common alleles exist that possess a core repeat unit of GATA. These include an allele with five repeats and consecutive alleles ranging from 4 to 17 repeat units in length. PCR products from Promega STR kits are 31 bp larger than those generated with Applied Biosystems kits for equivalent alleles.

A point mutation (T→A) 38 bp downstream of the STR repeat impacts the reverse primers for both the Applied Biosystems and Promega primer sets (see Figure 5.1). Applied Biosystems added an extra or "degenerate" unlabeled primer in their COfiler, SGM Plus, and Identifiler kits so that both possible alleles could be amplified (Wallin et al. 2002). On the other hand, Promega altered their D16S539 reverse primer sequence between kits but kept the overall amplicon size the same (Butler et al. 2001, Masibay et al. 2000, Krenke et al. 2002). The 3′-end of the PowerPlex 1.1 reverse primer was lengthened by five nucleotides to create the PowerPlex 16 reverse primer and thus move the primer mismatch caused by this mutation further into the primer to prevent allele dropout (Nelson et al. 2002, Krenke et al. 2002).

D18S51 is a simple tetranucleotide repeat found on the long arm of chromosome 18. It has a repeat motif of AGAA. A number of x.2 allele variants exist due to a 2 bp deletion from a loss of AG in the 3′-flanking region (Barber & Parkin 1996). More than 70 alleles have been reported for D18S51 making it one of the more polymorphic of the 13 core loci. PCR products from Promega's PowerPlex 2.1 STR kit are 22 bp larger than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for D18S51 (Butler et al. 2001, Krenke et al. 2002).

D21S11 is a complex tetranucleotide repeat found on the long arm of chromosome 21. A variable number of TCTA and TCTG repeat blocks surround a constant 43 bp section made up of the sequence {$[TCTA]_3$ TA $[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA}. The x.2 microvariant alleles arise primarily from a 2 bp (TA) insertion on the 3′-end of the repeat region (Brinkmann et al. 1996). PCR products from Promega's PowerPlex 2.1 STR kit are 17 bp larger than those generated with Applied Biosystems AmpFlSTR kits for equivalent alleles. The PowerPlex 2.1 and PowerPlex 16 STR kits contain the same PCR primers for D21S11.

Early papers in the literature by the Forensic Science Service had alleles named based on the dinucleotide subunit CV, where the V represents either an A, T, or G (Urquhart et al. 1994), while other authors adopted a different allele naming scheme based on the primary tetranucleotide repeat (Möller et al. 1994). As outlined in the European DNA Profiling Group interlaboratory study on D21S11 (Gill et al. 1997), a simple formula can be used to convert the Urquhart (U) designation into the Möller (M) equivalent:

$$M = \frac{1}{2} \times (U - 5)$$

Today most laboratories use the Moller allele notation since it fits the ISFG allele designation recommendation (Bär et al. 1997) format that is described later in this chapter.

D21S11 is far more polymorphic than can be easily detected with sized-based length separations. A careful search of the literature has revealed 90 reported alleles, many of which are the same length (see Appendix I). Fine differences in the D21S11 allele structures can only be determined by DNA sequencing since so many of the alleles have the same length but different internal sequence structure because some of the repeat units are switched around. For example, there are four different alleles designated as 30 repeats, which are indistinguishable by size-based methods alone (Appendix 1).

Additional Core and Common Loci Present in STR Kits

The STR markers described in this section are present in commercial STR typing kits and thus analyzed in conjunction with the core (required) loci covered in the previous section. Because these markers are included in the STR multiplexes co-amplified with core loci used for developing DNA databases, potentially millions of DNA samples will be run with these loci as DNA databases grow worldwide.

In 1999, Applied Biosystems released the AmpFlSTR SGM Plus kit that co-amplifies 10 STR loci including two additional STRs: D19S433 and D2S1338. With the adoption of the SGM Plus kit by the Forensic Science Service and much of Europe, the number of population data on the STR loci D19S433 and D2S1338 is now in the millions of profiles. These two loci are also part of the Identifiler and NGM 16plex STR kits as well as the PowerPlex ESX 17, PowerPlex ESI 17, and PowerPlex 18D.

Likewise, the Promega Corporation included two pentanucleotide STR loci, Penta E and Penta D, in their PowerPlex 2.1 and PowerPlex 16 kits that were released at the turn of the century. Since the German national DNA database requires analysis of the complex hypervariable STR locus SE33, commercial kits are available with SE33 including PowerPlex ES, PowerPlex ESX 17, PowerPlex ESI 17, SEfiler Plus, and NGM SElect.

D2S1338 is a compound tetranucleotide repeat found on the long arm of chromosome 2. Alleles ranging from 10 to 31 repeats have been observed. D2S1338 has a high heterozygosity and is present in the SGM Plus, Identifiler, SEfiler Plus, MiniFiler, NGM, and NGM SElect kits from Applied Biosystems and the PowerPlex ESI 16/17, PowerPlex ESX 16/17, and PowerPlex 18D kits from Promega.

D19S433 is a compound tetranucleotide repeat located on chromosome 19 with observed alleles ranging from 5.2 to 20 repeats. The x.2 alleles are due to an "AG" deletion prior to the core AAGG repeat (Heinrich et al. 2005). D19S433 is present in the SGM Plus, Identifiler, SEfiler Plus, NGM, and NGM SElect kits from Applied Biosystems and the PowerPlex ESI 16/17, PowerPlex ESX 16/17, and PowerPlex 18D kits from Promega.

Penta D is a pentanucleotide repeat found on chromosome 21 about 25 Mb from D21S11. It is present in the PowerPlex 16 and PowerPlex 18D kits from Promega.

Penta E is a pentanucleotide repeat with very low stutter product formation that is located on the long arm of chromosome 15 with alleles ranging from 5 to 32 repeats. It is present in the PowerPlex 2.1, PowerPlex 16, and PowerPlex 18D kits from Promega.

D1S1656 is a tetranucleotide repeat found on the long arm of chromosome 1 with alleles ranging from 8 to 20.3 repeats. The x.3 alleles arise from a TGA insertion typically after four full TAGA repeats. It is part of the extended European Standard Set and possesses a high heterozygosity.

D2S441 is a tetranucleotide repeat located on the short arm of chromosome 2 more than 60 Mb from TPOX. It can be amplified as a miniSTR and works well on degraded DNA samples (Coble & Butler 2005). Some x.3 alleles have been observed as well as same-size, different-sequence alleles (Phillips et al. 2011). D2S441 is part of the extended European Standard Set and is present in NGM and NGM SElect kits from Applied Biosystems and the PowerPlex ESI and ESX Systems from Promega.

D10S1248 is a simple tetranucleotide repeat found on the long arm of chromosome 10 and possesses 7 to 19 GGAA repeats. It can be amplified as a miniSTR and works well on degraded DNA samples (Coble & Butler 2005). It is part of the extended European Standard Set and is present in NGM and NGM SElect kits from Applied Biosystems and the PowerPlex ESI and ESX Systems from Promega.

D12S391 is a highly polymorphic compound tetranucleotide found on the short arm of chromosome 12 only 6.3 megabases from vWA. It possesses over 50 different alleles ranging from 13 to 27.2 repeats in length. A number of same-size, different-sequence alleles have been identified through sequence analysis (Lareu et al. 1996, Phillips et al. 2011). It is part of the extended European Standard Set and is present in NGM and NGM SElect kits from Applied Biosystems and the PowerPlex ESI and ESX Systems from Promega.

D22S1045 is a simple trinucleotide repeat found on chromosome 22 with alleles ranging from 7 to 20 ATT repeats. While it is not as polymorphic as most of the other 23 core and common STR loci, it can be amplified as a miniSTR and works well on degraded DNA samples (Coble & Butler 2005). D22S1045 is part of the extended European Standard Set and is present in NGM and NGM SElect kits from Applied Biosystems and the PowerPlex ESI and ESX Systems from Promega.

SE33 is the most variable STR locus studied to date. It is located on the long arm of chromosome 6 and contains a core AAAG repeat structure. Appendix 1 describes 178 observed alleles ranging from 3 to 49 repeats. Sequence analysis has revealed 15 different 29.2 alleles possessing a variety of internal sequence combinations. SE33 is a core locus for the German national DNA database and with its renewed availability in the NGM SElect and PowerPlex ESI 17 and ESX 17 Systems is being adopted by other laboratories around Europe.

Other Non-Kit Based Common STR Loci

The 13 core loci used within the United States for CODIS and the additional STRs present in the STR kits (particularly the extended ESS loci) are effective DNA markers for human identification and will most likely continue to be used for some time especially due to the existence now of large DNA databases containing information from these autosomal STR loci (see Chapter 8). However, these 23 markers are by no means the only STRs that have been evaluated or used by forensic labs around the world. Dozens of other markers have been used, some quite extensively (Grubwieser et al. 2007).

For example, a 26plex assay developed by researchers at the National Institute of Standards and Technology (Hill et al. 2009) enables collection of information from 15 chromosomes with 25 STR markers (and amelogenin for sex-typing) that are sufficiently spaced from current core STR loci to be considered unlinked and thus enable the use of the product rule when combining information across the various loci (see Table 10.3). This 26plex assay

has been used in conjunction with Identifiler to provide 40 STRs in order to strengthen relationship comparisons (O'Connor et al. 2010).

In addition to the autosomal STR loci described in this chapter, both kit and non-kit Y-chromosome STR markers (Chapter 13) and X-chromosome STR loci (Chapter 15) are covered in this book along with their applications in forensic investigations and relationship testing. Cat and dog STR loci are also described in Chapter 16 (see Table 16.1 and Table 16.2).

STR ALLELE NOMENCLATURE

To aid in interlaboratory reproducibility and comparisons of data, a common nomenclature has been developed in the forensic DNA community. DNA results cannot be effectively shared unless all parties are speaking the same language and referring to the same conditions. (It would do little good to describe the recipe for baking a cake in a language that is not understood by both the recipe giver and the chef. For example, if the recipe says to turn the oven on to 450 degrees Fahrenheit and the chef uses 450 Kelvin [≈250°F], the results would be vastly different.)

If one laboratory calls a sample 15 repeats at a particular STR locus and the same sample is designated 16 repeats by another laboratory, a match would not be considered, and the samples would be assumed to come from separate sources. As discussed in Chapter 8, the advent of national DNA databases with many laboratories contributing information to those databases has made it crucial to have internationally accepted nomenclature for designating STR alleles.

A repeat sequence is named by the structure (base composition) of the core repeat unit and the number of repeat units. However, because DNA has two strands, which may be used to designate the repeat unit for a particular STR marker, more than one choice is available and confusion can arise without a standard format. Also, where an individual starts counting the number of repeats can also make a difference. With double-stranded DNA sequences being read in the 5′ to 3′ direction, the choice of the strand impacts the sequence designation. For example, the "top" strand for an STR marker may be 5′-…$(GATA)_n$…-3′

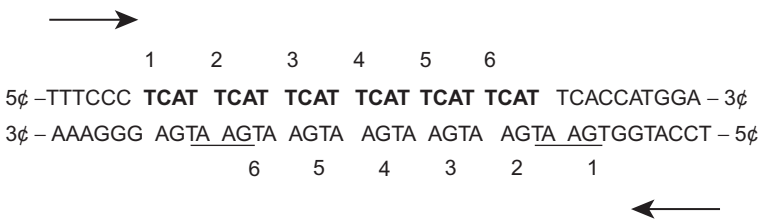


FIGURE 5.6 Example of the DNA sequence in a STR repeat region. Note that using the top strand versus the bottom strand results in different repeat motifs and starting positions. In this example, the top strand has 6 TCTA repeat units while the bottom strand has 6 TGAA repeat units. Under ISFG recommendations, the top strand from GenBank should be used. Thus, this example would be described as having [TCAT] as the repeat motif. Repeat numbering, indicated above and below the sequence, proceeds in the 5′-to-3′ direction as illustrated by the arrows.

while the "bottom" strand for the same sequence would be 5′-…$(TATC)_n$ …-3′. Depending on the sequence surrounding the repeat region, the core repeat could be shifted relative to the other strand (Figure 5.6).

ISFG DNA Commission Recommendations

The International Society of Forensic Genetics (ISFG), which was founded in 1968 and formerly known as the International Society of Forensic Haemogenetics (ISFH), today represents a group of approximately 1100 scientists from more than 60 countries (see http://www.isfg.org/). Meetings are held biannually to discuss the latest topics in forensic genetics. Every few years, as a specific need arises, a DNA Commission of the ISFG is formed and makes recommendations on the use of genetic markers. These publications are available at http://www.isfg.org/Publications/DNA+Commission and include guidelines for designating STR allele nomenclature:

- Naming variant alleles (Bär et al. 1994)
- Repeat nomenclature (Bär et al. 1997)
- Y-STR use in forensic analysis (Gill et al. 2001)
- Additional Y-STRs—nomenclature (Gusmão et al. 2006)

The ISFG 1994 recommendations focused on allelic ladders and designation of alleles that contain partial repeat sequences. The ISFG 1997 guidelines discussed the sequence and repeat designation of STRs. Further comments on STR allele nomenclature with specific interest in Y-chromosome markers were published in 2001 (Gill et al. 2001) and 2006 (Gusmão et al. 2006). A more recent detailed description of STR allele nomenclature rules with specific examples from Y-chromosome loci was released by scientists from the U.S. National Institute of Standards and Technology (NIST) primarily to help the genetic genealogy community (Butler et al. 2008).

The 1994 ISFG DNA Commission publication addressed designations of alleles containing partial repeat sequences: "When an allele does not conform to the standard repeat motif of the system in question it should be designated by the number of complete repeat units and the number of base pairs of the partial repeat. These two values should be separated by a decimal point" (Bär et al. 1994). For example, an allele with $[AATG]_5$ ATG $[AATG]_4$ is designated as a "9.3" since it contains nine full AATG repeats plus three additional nucleotides. Thus, tetranucleotide repeats (i.e., those containing four nucleotides in the repeat motif) could have x.1, x.2, and x.3 variant alleles that exhibit one, two, or three additional nucleotides beyond the number of complete repeat units found in the allele.

The 1997 ISFG recommendations are reviewed below (Bär et al. 1997):

Choice of the Strand

- For STRs within protein coding regions (as well as in the intron of the genes), the coding strand should be used. This would apply to STRs such as vWA (GenBank: M25716), TPOX (GenBank: M68651), and CSF1PO (GenBank: X14720).
- For repetitive sequences without any connection to protein coding genes like many of the D#S### loci, the sequence originally described in the literature of the first public database

entry shall become the standard reference (and strand) for nomenclature. Examples here include D18S51 (GenBank: L18333) and D21S11 (GenBank: M84567).
- If the nomenclature is already established in the forensic field but not in accordance with the aforementioned guideline, the nomenclature shall be maintained to avoid unnecessary confusion. This recommendation applies to the continued use by some laboratories of the "AATG repeat" strand for the STR marker TH01. The GenBank sequence for TH01 uses the coding strand and therefore contains the complementary "TCAT repeat" instead.

Choice of the Motif and Allele Designation

- **The repeat sequence motif should be defined so that the first 5′-nucleotides that can define a repeat motif are used.** For example, 5′-GG TCA TCA TCA TGG-3′ could be seen as having 3 × TCA repeats or 3 × CAT repeats. However, under the recommendations of the ISFH committee only the first one (3 × TCA) is correct because it defines the first possible repeat motif.
- Designation of incomplete repeat motifs should include the number of complete repeats and, separated by a decimal point, the number of base pairs in the incomplete repeat. Examples of "microvariants" with incomplete repeat units include allele 9.3 at the TH01 locus. TH01 allele 9.3 contains nine tetranucleotide AATG repeats and one incomplete ATG repeat of three nucleotides (Puers et al. 1993). Another microvariant example is allele 22.2 at the FGA locus, which contains 22 tetranucleotide repeats and one incomplete repeat with two nucleotides (Barber et al. 1996).
- Allelic ladders containing sequenced alleles that are named according to the recommendations listed above should be used as a reference for allele designation in unknown samples. Allelic ladders may be commercially obtained or prepared in house and should contain all common alleles.

This article further notes: "For those situations where two or more nomenclatures already exist, priority should be given to the nomenclature that more closely adheres to the [1997 ISFG] guidelines. If this is not possible, priority shall be given to the nomenclature that was documented first" (Bär et al. 1997). When reviewing the STR literature prior to 1997, an individual should keep in mind that repeat nomenclatures often differ from the ISFG 1997 guidelines. This fact can lead to some confusion if one is not careful. For example, early descriptions of the STR locus TH01 by the Forensic Science Service label the repeat TCAT (Kimpton et al. 1993) while Caskey and co-workers described the TH01 repeat as AATG (Edwards et al. 1991).

The 2006 ISFG DNA Commission made the following eight nomenclature recommendations (Gusmão et al. 2006):

1. Alleles should be named according to the total number of contiguous variant and non-variant repeats determined from sequence data. Single repeat units located adjacent to the main repeat array and consisting of the same sequence as the main variable repeat should be considered as part of the repeat motif. For example, a hypothetical STR allele with the sequence …$(GATA)_n(GACA)_2(GATA)$… should be considered to have $n + 2 + 1$ repeats.

2. Repetitive motifs that are not adjacent to the variable stretch and have three or less units and show no size variation within humans or between humans and chimpanzees should not be included in the allele nomenclature. For example, a hypothetical STR with the sequence … … $(GATA)_n(GACA)_2N_8(GATA)_3$…, where N contains nucleotides that are not part of the repeat motif, should be called $n + 2$, excluding the non-adjacent $(GATA)_3$ repetitive stretch from the allele nomenclature. If the number of interrupting nucleotides (N) is similar to or less than the number of nucleotides in the repeat motif, then the region is considered one repeat unit with a length corresponding to the total number of nucleotides. Thus, …$(GATA)_n(GACA)_2N_4(G$ $ATA)_3$… is considered as one complex locus with $n + 2 + 1 + 3$ units, while …$(GATA)_n(GA$ $CA)_2N_5(GATA)_3$… is considered to be two loci with $n + 2$ and 3 units, respectively, of which $n + 2$ would be included in the primary STR allele nomenclature.
3. Intermediate alleles (e.g., 11.1) fall into two classes: an insertion/deletion either (a) within the repeat motif or (b) in the flanking region encompassed by the PCR primer positions. If the partial repeat is found within the repeat motif, such as …$(GATA)_nT(GATA)_m$, alleles should be called as noted in the 1994 ISFG recommendations: "…by the number of complete repeat units and the number of base pairs of the partial repeat separated by a decimal point" (Bär et al. 1994).
4. Intermediate alleles arising due to mutations in the flanking sequences that alter the length or electrophoretic migration of a PCR product should be designated by additional information indicated after the number of complete STR repeat units. For example, an allele with 11 repeats and a T insertion at nucleotides 40 upstream from the repeat is not named "11.1" but rather "11(U40Tins)" where 11 stands for the number of complete repeats, U40 indicates the direction and position of the mutation relative to the STR repeat block (i.e., the mutation is located 40 bases upstream of the repeat), and "Tins" indicates that a T nucleotide has been inserted. If the exact position of the deletion or insertion cannot be determined because it is part of a homopolymeric tract (i.e., a stretch of the same nucleotides such as TTTTT), then the deletion or insertion should be assigned to the highest numbered end of the homopolymeric stretch. Using Gusmão et al. 2006 Figure 2 as an example, the deletion that gives rise to the "16.3" allele should more appropriately be referred to as a "17D80Tdel" allele since the single T deletion occurs at the end of a polymeric T stretch that is 80 nucleotides downstream of the repeat region.
5. Point mutations in a PCR primer binding region may prevent sufficient annealing of this primer and result in a "null" or "silent" allele due to failure to generate a detectable amount of PCR product. It is recommended that point mutations which impact primer annealing be verified by DNA sequence analysis and published using a designation as in recommendation #4. For example, DYS438 (D7A→C) would indicate that the "A" nucleotide 7 bases downstream of the DYS438 repeat has changed into a "C" nucleotide in the tested STR allele.
6. If no additional sequence variation is found in the 166 Y-STR markers described by Kayser et al. 2004, then these authors' locus delimitation criteria should be adopted.
7. Journal editors, reviewers, and organizers of quality assurance schemes should focus on the use of standardized nomenclatures in order to obtain uniformity and avoid the spread of confusing nomenclatures.
8. Commercial Y-STR kits should follow the nomenclature recommendations so that direct comparisons between results obtained with different kits are possible.

While these guidelines provide a framework for STR allele nomenclature designation, they do not capture every possible permutation that exists, particularly with complex repeats. Following recommendations #1 and #2 described above, scientists at NIST devised what they termed the "one-change-rule" in that a single change to the repeat motif can be allowed in deciding what to include or not in an STR repeat block (Butler et al. 2008). However, when the single change in the repeat motif creates an adjacent homopolymeric stretch, it was decided not to include this portion in the repeat count. For example, with the repeat motif of CTT, if an adjacent sequence of TTT occurs (e.g., DYS481), then this approach only counts the number of CTT repeats. On the other hand, with a repeat structure of $(GATA)_n(GACA)$, the repeat count would be $n + 1$.

Today most forensic DNA scientists do not worry about STR allele nomenclature because STR typing is almost universally performed with commerically available kits containing allelic ladders prepared with alleles designated according to the ISFG rules by the manufacturer. However, occasionally there are differences in nomenclature that need to be spelled out to avoid confusion, such as with the Y-STR locus GATA-H4 contained in the Yfiler kit (Mulero et al. 2006). The use of certified reference materials and common positive controls with known genotypes also further promotes consistency across laboratories in terms of STR allele designation.

Most of the time measured alleles will correlate to the alleles present in the STR kit allelic ladders and be appropriately designated as "on-ladder" alleles using genotyping software. However, there are exceptions, and new alleles are being discovered as more DNA samples are examined around the world. In addition, sometimes more than two alleles can be inherited or exhibited as is the case with the so-called triallelic patterns. Finally, due to variation in the STR flanking regions, some PCR primers may fail to amplify a particular allele—a situation known as allele dropout due to a silent or "null" allele. These topics are discussed in the following sections.

NULL (SILENT) ALLELES

When amplifying DNA fragments that contain STR repeat regions, it is possible to have a phenomenon known as *allele drop-out*. Sequence polymorphisms are known to occur within or around STR repeat regions. These variations can occur in three locations (relative to the primer binding sites): within the repeat region, in the flanking region, or in the primer-binding region.

If a base pair change occurs in the DNA template at the PCR primer binding region, the hybridization of the primer can be disrupted resulting in a failure to amplify, and therefore failure to detect an allele that exists in the template DNA. More simply, the DNA template exists for a particular allele but fails to amplify during PCR due to primer hybridization problems. This phenomenon results in what is known as a *null allele*. Fortunately null alleles are rather rare because the flanking sequence around STR repeats is fairly stable and consistent between samples.

Concordance Studies Aid Discovery of Null Alleles

Null alleles have been "discovered" by the observation of different typing results when utilizing independent STR primer sets. During a comparison of STR typing results on

600 population samples at the vWA locus, one sample typed 16,19 with Promega's PowerPlex kit and 16,16 with Applied Biosystem's AmpFlSTR Blue kit (Kline et al. 1998). In this case, the vWA allele 19 dropped out with the AmpFlSTR vWA primer set due to a sequence polymorphism near the 3′-end of the forward primer (Walsh 1998).

Allele dropout may occur due to mutations (variants) at or near the 3′-end of a primer. If a primer binding site mutation exists, then little or no primer extension will occur during PCR. Of course, this failure to amplify depends on the PCR conditions including the annealing temperature used. In the situation described above, the vWA allele 19 was present in the sample but failed to be amplified by one of the primer sets. It was later reported that the null allele resulted from a rare A→T nucleotide change in the DNA template that occurred at the second base from the 3′-end of the AmpFlSTR vWA forward primer (Walsh 1998).

Potential null alleles resulting from allele dropout can be predicted by statistical analysis of the STR typing data. The observed number of homozygotes can be compared to the expected number of homozygotes based on Hardy-Weinberg equilibrium (Chakraborty et al. 1992). An abnormally high level of homozygotes would indicate the possible presence of null alleles. Thus, each set of population data should be carefully examined when new STR markers are being tested in a forensic DNA laboratory.

A number of primer concordance studies have been conducted in the past few years as use of various STR kits has become more prevalent. An examination of over 2000 samples comparing the PowerPlex 16 kit to the Profiler Plus and COfiler kit results found 22 examples of allele dropout due to a primer mismatch at 7 of the 13 core STR loci: CSF1PO, D8S1179, D16S539, D21S11, FGA, TH01, and vWA (Budowle et al. 2001, Budowle & Sprecher 2001).

Use of Degenerate Primers in Commercial Kits

In some cases, STR kit manufacturers have added an additional PCR primer to the assay that can hybridize properly to the alternative allele when it exists in a sample. This has been the preferred solution for Applied Biosystems (e.g., Wallin et al. 2002) while Promega has moved their primers to overcome allele dropout problems (e.g., Nelson et al. 2002). According to their publications, Applied Biosystems has added an additional primer to correct for single point mutations in AmpFlSTR primer binding sites for D16S539 (Wallin et al. 2002), vWA (Lazaruk et al. 2001), and D8S1179 (Leibelt et al. 2003).

STR TYPING

As described in the next chapter, STR alleles are separated and sized using capillary electrophoresis. Data interpretation software is used to size PCR products amplified using STR primers and to convert the DNA sizes to STR alleles through use of allelic ladders (Figure 5.7). As noted earlier in this chapter, allelic ladders are provided by the commercial kit providers (although a laboratory could prepare its own ladder by gathering alleles from a diverse set of samples). The ladders are prepared with commonly observed STR alleles usually spaced a single repeat unit apart from one another. The ladder alleles are PCR-amplified with the same primers as provided in the STR typing kit for testing unknown samples. Thus, samples amplified with a kit will produce alleles that are the same size as an allele in the allelic ladder.

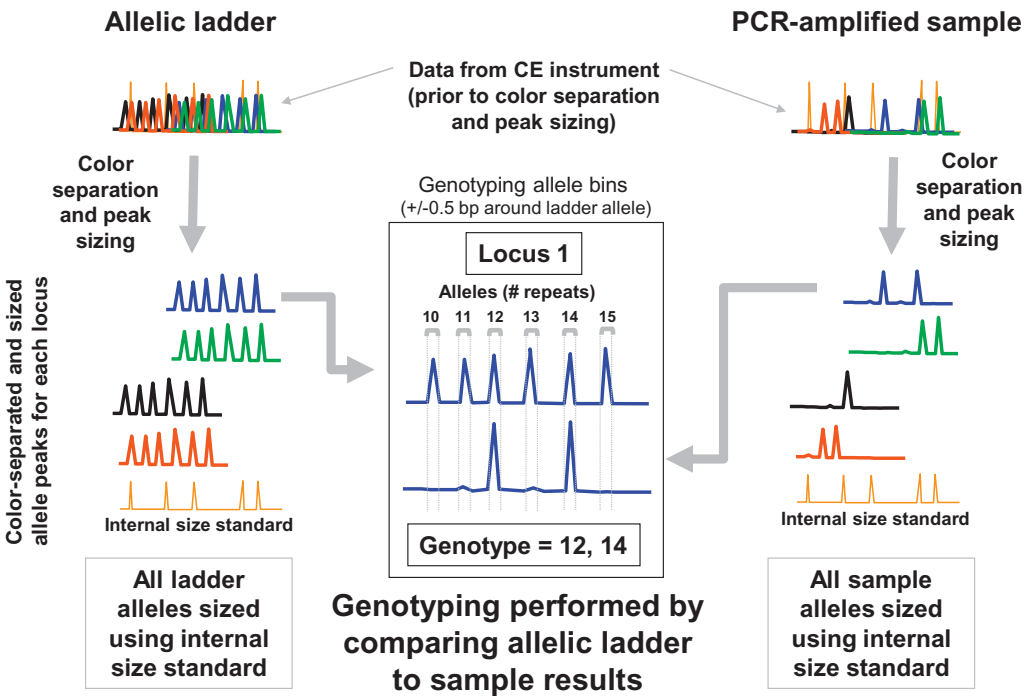


FIGURE 5.7 Genotyping is performed through a comparison of sized peaks from PCR-amplified samples to allele size bins. These allele bins are defined with the genotyping software using size information from an allelic ladder run with each batch of samples. Any peak falling in a particular dye color and allele bin size range is designated as an allele for that locus. Peaks in both the allelic ladder and the PCR-amplified samples are sized using the same internal size standard so that they may be compared to one another.

Each allele or "rung" of an allelic ladder has been characterized in terms of the number of repeats it contains through DNA sequencing. The STR kit manufacturer supplies information to the users regarding the allele nomenclature and number of repeats present in each allele in the ladder. The data analysis software enables a conversion of all peaks in samples being processed from DNA size (relative to a common internal size standard) to repeat number. The DNA sizes of the allelic ladder alleles are used to calibrate size ranges for allele classification. A common size range for the genotyping allele bins is ±0.5bp around each allele. This size range enables PCR products that are 1bp different from one another to be differentiated. Due to slight changes in instrument environmental conditions over time, allelic ladders are run regularly (typically with every batch of samples) in order to keep the size-to-allele conversion process well calibrated.

Any peak with the same dye-color label that is sized within half a base pair of an allelic ladder is designated by the STR genotyping software as being that allele. An analyst while performing DNA profile review assesses whether or not the labeled peak is truly an allele or an artifact, such as a stutter product or pull-up of signal from another dye color. Further information on artifacts and data interpretation will be covered in the forthcoming *Advanced Topics in Forensic DNA Typing: Interpretation*.

Same Length but Different Sequence Alleles

Complex repeat sequences, such as those found in D21S11, can contain variable repeat blocks in which the order is switched around for alleles that are the same length. For example, the STR locus D21S11 has four alleles that are all 210 bp when amplified with the Identifiler kit (Appendix 1). While these alleles would be sized based on overall length to be "allele 30," they contain repeat blocks of 4-6-CR-12, 5-6-CR-11, 6-5-CR-11, and 6-6-CR-10 for the pattern [TCTA]-[TCTG]-constant region (CR)-[TCTA]. In such cases, variant alleles would only be detectable with complete sequence analysis.

It is important to realize that from an operational point of view internal allele variation is not significant. In the end a match is being made against many loci not just one, such as D21S11, with possible internal sequence variation. Most of the STR loci used in human identity testing have not exhibited internal sequence variation, particularly the simple repeat loci TPOX, CSF1PO, D5S818, D16S539, TH01, D18S51, and D7S820. Alleles are binned based on measured size with STR typing since sequence analysis of individual alleles is too time consuming and, because STR variation is primarily size-based, would rarely reveal additional information.

Over the past several years, work with mass spectrometry has demonstrated that same-size STR alleles with different internal sequences or alleles with sequence variation in the flanking region can be resolved from one another using a base composition approach (Oberacher et al. 2008, Pitterl et al. 2008, Planz et al. 2009, Pitterl et al. 2010). This sequence variation is sometimes refered to as a SNPSTR, where single nucleotide variation is coupled with STR allele identification. As seen in Appendix 1, certain STR loci, such as SE33, are more prone to internal sequence variation.

A comparison of the variation observed in 11 STRs—the 10 SGM Plus loci and SE33—in DNA samples from 94 Yakut and 108 Khoisan individuals was performed using conventional STR analysis, mass spectrometric STR analysis, and direct sequencing (Pitterl et al. 2010). The mass spectrometry approach that enabled both sequence and length variation to be detected clearly expanded the number of detectable alleles.

These same length but different internal sequence alleles can be used in research to track mutations if the sequences can be differentiated from one another. A C→T polymorphism was discovered 13 nucleotides upstream of the D5S818 repeat region (Edwards & Allen 2004). A restriction enzyme cutting site is created when the T is present that enables differentiation of samples with and without the polymorphism. Use of this SNPSTR assay permitted evaluation of mutations that were observed of which 34 out of 40 were paternal in origin and 23 of 26 (where the repeat change could be determined) were single repeat unit shifts. This study also found that while alleles 13 and 14 were prone to mutation, allele 11 was not—suggesting that not all alleles have equal frequencies of mutation (Edwards & Allen 2004). In fact, the American Association of Blood Banks (AABB) annual report on relationship testing laboratories is now seeking to quantify these allele-specific mutation events (AABB 2008).

GENDER IDENTIFICATION WITH AMELOGENIN

The ability to designate whether a sample originated from a male or a female source is useful in sexual assault cases, where distinguishing between the victim and the perpetrator's

evidence is important. Likewise, missing persons and mass disaster investigations can benefit from gender identification of the remains. Over the years a number of gender identification assays have been demonstrated using PCR methods (Sullivan et al. 1993, Eng et al. 1994, Reynolds & Varlaro 1996). By far the most popular method for sex-typing today is the amelogenin system as it can be performed in conjunction with STR analysis.

Amelogenin is a gene that codes for proteins found in tooth enamel. The British Forensic Science Service was the first to describe the particular PCR primer sets that are used so prevalently in forensic DNA laboratories today (Sullivan et al. 1993). These primers flank a 6bp deletion within intron 1 of the amelogenin gene on the X homologue (Figure 5.8). PCR amplification of this area with their primers results in 106bp and 112bp amplicons from the X and Y chromosomes, respectively. Primers, which yield a 212bp X-specific amplicon and a 218bp Y-specific product by bracketing the same 6bp deletion, were also described in the original amelogenin paper (Sullivan et al. 1993) and have been used in conjunction with the D1S80 VNTR system (Budowle et al. 1996).

An advantage with the above approach, i.e., using a single primer set to amplify both chromosomes, is that the X chromosome product itself plays a role as a positive control. This PCR-based assay is extremely sensitive. Mannucci and co-workers were able to detect as little as 20 pg (≈3 diploid copies) as well as sample mixtures where female DNA was in 100-fold excess of male DNA (Mannucci et al. 1994).

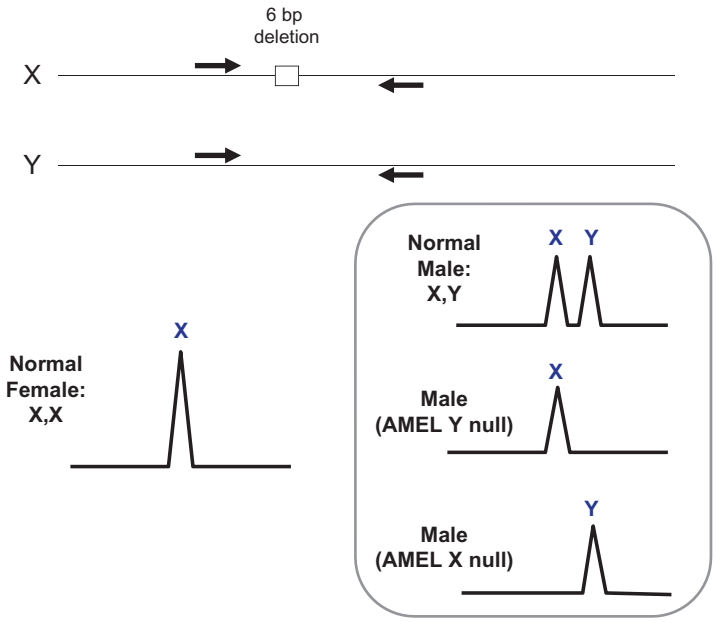


FIGURE 5.8 Schematic of the amelogenin sex-typing assay. The X and Y chromosomes contain a high degree of sequence homology at the amelogenin (AMEL) locus and primers, as depicted by the arrows, can target a 6bp deletion that is present only on the X chromosome (Sullivan et al. 1993). In most circumstances, the presence of a single X peak indicates that the sample comes from a female while two peaks identifies the sample's source as male. Both AMEL Y and AMEL X null alleles have been reported.

Other regions of the amelogenin gene have size differences between the X and Y homologues and may be exploited for sex-typing purposes. For example, Eng and co-workers (1994) used a single set of primers that generated a 977bp product for the X-chromosome and a 788bp fragment for the Y-chromosome. In this case, a 189bp deletion in the Y relative to the X-chromosome was used to differentiate the two chromosomes.

A careful study found that 19 regions of absolute homology, ranging in size from 22 bp to 80 bp, exist between the human amelogenin X and Y genes that can be used to design a variety of primer sets (Haas-Rochholz & Weiler 1997). Thus, by spanning various deletions of the X and/or Y chromosome, it is possible to generate PCR products from the X and Y homologues that differ in size and contain size ranges that can be integrated into future multiplex STR amplifications.

While amelogenin is an effective method for sex-typing biological samples in most cases, the results are not foolproof either due to primer binding sites that lead to null alleles or chromosomal deletions. Amelogenin Y allele dropouts have been observed due to loss of portions of the Y chromosome in some population groups. Amelogenin X allele dropouts have been seen primarily due to primer binding site mutations.

Amelogenin Y Allele Dropout

A rare deletion of the amelogenin gene on the Y chromosome can cause the Y chromosome amplicon to be absent (Santos et al. 1998). In such a case, a male sample would falsely appear as a female with only the amelogenin X allele being amplified. It appears that this deletion of the Y chromosome amelogenin region is more common in Indian populations (Thangaraj et al. 2002) than those of European or African origins. A study of almost 30,000 males in the Austrian National DNA database revealed that only six individuals lacked the amelogenin Y-amplicon (Steinlechner et al. 2002). These individuals were verified to be male with Y-STRs and amplification of the SRY region (see Chapter 13).

More recent studies have attempted to map the Y deletions in detail and to track the specific biogeographic ancestry of these interesting variants (Cadenas et al. 2007, Jobling et al. 2007). Through examining adjacent STR markers and sequence-specific tag sites, the extent of the Y chromosome deletion can be mapped (Takayama et al. 2009). When amplifying Y-STR loci, the locus DYS458 in Yfiler (see Chapter 13) is the most likely one to be lost with amelogenin Y deletions due to its close proximity to the AMEL Y (see Figure 13.6).

Amelogenin X Allele Dropout

Amelogenin X allele dropout has also been observed in males. In this case only the amelogenin Y-amplicon is present (Shewale et al. 2000, Alves et al. 2006, Maciejewska & Pawłowski 2009). In one study, this phenomenon was observed only three times out of almost 7000 males examined (Shewale et al. 2000). The authors of this study felt that the AMEL X null was most likely a result of a rare polymorphism in the primer binding sites for the amelogenin primers used in commercial STR kits. A different set of amelogenin primers targeting the same 6bp deletion on the X chromosome amplified both the X and Y alleles of amelogenin (Shewale et al. 2000). However, in some populations, this loss of the AMEL X allele is more common. In a study of 503 individuals from São Tomé Island (West Africa),

10 male individuals displayed only the Y allele from amelogenin amplification due to a primer binding site mutation in the AMEL X allele (Alves et al. 2006). A different mutation caused one male out of 5534 Polish males tested to display only the AMEL Y allele (Maciejewska & Pawłowski 2009).

A report of males examined in paternity testing labs using Applied Biosystems STR typing kits found that there were a higher number of African American males showing only the AMEL Y allele (i.e., the AMEL X allele dropped out). Still, this AMEL X null is fairly rare being seen only 48 times in 144,391 males tested or 0.03% of the time (AABB 2008).

STRBASE: AN INTERNET RESOURCE

The rapid growth of the human identification applications for STR loci ensures that static written materials, such as this book, will quickly become out-of-date. New alleles are constantly being discovered (including "off-ladder" microvariant alleles), additional STR markers are being developed, and population data increases with each month of published journals. Indeed, a growing list of publications describing the application of STR loci to forensic DNA typing has exceeded 3500 references.

The World Wide Web enables dynamic sources of information to be widely available. More than a decade ago a website was created to enable forensic scientists to keep abreast of the rapidly evolving field of DNA typing. STRBase was officially launched in July 1997 and is maintained by the Applied Genetics Group of the National Institute of Standards and Technology (Butler & Reeder 1997, Ruitberg et al. 2001, Butler et al. 2008). STRBase may be reached via the World Wide Web using the following URL: http://www.cstl.nist.gov/biotech/strbase.

STRBase contains a number of useful sections. Continually updated information includes the listing of references related to STRs and DNA typing (over 3500 references), addresses for scientists working in the field, and new microvariant or "off-ladder" STR alleles. Other information that is updated less frequently includes STR fact sheets (with allele information similar to Appendix 1), links to other web pages, a review of technology used for DNA typing as well as published primer sequence information, population data for STR markers, and summaries of NIST forensic DNA reference materials. Sections with information on Y-STR markers and low template DNA testing are also available.

STR markers have become important tools for human identity testing. Commercially available STR kits are now widely used in forensic and paternity testing laboratories. The adoption of the 13 CODIS core loci for the U.S. national DNA database ensures that these STR markers will be used for many years to come.

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